Type MS · The discriminant: equal, real or no real roots

Answer 1 of 12 · Question 1 part whole · 4 marks

2022 JUN · Paper 02 · find part whole in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · The discriminant: equal, real or no real roots

Answer 2 of 12 · Question 1 part whole · 5 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

1	$b^2-4ac>0 \Rightarrow 8^2-4\times k\times 3k>0$ $64-12k^2>0$ cvs: $k=\pm\sqrt{\frac{64}{12}}=\left[\pm\frac{4\sqrt{3}}{3}\right]$ or $\left[\pm\frac{4}{\sqrt{3}}\right]$ $\Rightarrow -\frac{4\sqrt{3}}{3}<k<\frac{4\sqrt{3}}{3}$	M1 M1A1 M1A1 [5]
		Total 5 marks

Mark	Notes
M1	Applies the correct values, with the correct inequality to $b^2-4ac>0$
M1	Attempts to find two critical values by solving the quadratic equation, which must be of the form $k^2=$ constant [oe] using a correct method. Accept as a minimum solution $k=\pm\sqrt{\text{constant}}$ Allow simplified or unsimplified. Ignore any inequalities, equal signs etc
A1	For the correct critical values simplified or unsimplified Award this mark for correct critical values. Ignore $=, <, >$ or even $\leqslant, \geqslant$ M0M1A1 is a possible marking pattern.
M1	Simplifies the critical values to the required form and writes down the **inside** region for **their TWO** critical values. If they solve a linear equation for k this mark is not available. Allow use of x for this mark and also allow $\leqslant$ in place of $<$.
A1	For the correct region specified correctly in either of the two forms specified in terms of k. That is, a continuous inside region. Accept $-\frac{4}{\sqrt{3}}<k<\frac{4}{\sqrt{3}}$ oe Accept also for example $k>-\frac{4}{\sqrt{3}}$ AND $k<\frac{4}{\sqrt{3}}$

Type MS · The discriminant: equal, real or no real roots

Answer 2 of 12 · Question 1 part whole · 5 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · The discriminant: equal, real or no real roots

Answer 3 of 12 · Question 7 part a · 4 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

7 (a)	$16-4k(2k-7)\geq 0$	M1
	$2k^2-7k-4\leq 0$	M1
	$(2k+1)(k-4)\leq 0$	M1
	$-\frac{1}{2}\leq k\leq 4$ Accept $-\frac{1}{2}<k<4$	A1 (4)
(b)	$\alpha+\beta=\frac{4}{k}$ $\alpha\beta=\frac{2k-7}{k}$	B1
	$\frac{\alpha+1}{\alpha}+\frac{\beta+1}{\beta}=\frac{2\alpha\beta+\alpha+\beta}{\alpha\beta}=\frac{2\left(\frac{2k-7}{k}\right)+\frac{4}{k}}{\frac{2k-7}{k}}$	M1 M1
	$\frac{4k-10}{k}\times\frac{k}{2k-7}=\frac{2(2k-5)}{2k-7}$	A1
	$\frac{\alpha+1}{\alpha}\times\frac{\beta+1}{\beta}=\frac{\alpha\beta+\alpha+\beta+1}{\alpha\beta}=\frac{\frac{2k-7}{k}+\frac{4}{k}+1}{\frac{2k-7}{k}}$	M1 M1
	$\frac{3k-3}{k}\times\frac{k}{2k-7}=\frac{3(k-1)}{2k-7}$	A1
	$(2k-7)x^2-2(2k-5)x+3(k-1)=0$	A1 (8)
	ALTERNATIVE METHOD	
	Let $w=\frac{x+1}{x}$	B1
	$x=\frac{1}{w-1}$	M1
	Hence $k\left(\frac{1}{w-1}\right)^2-4\left(\frac{1}{w-1}\right)+2k-7=0$	M1
	$\frac{k}{(w-1)^2}-\frac{4}{w-1}+2k-7=0$	A1
	$k-4(w-1)+(2k-7)(w-1)^2=0$	M1
	$k-4w+4+(2k-7)(w^2-2w+1)=0$	M1
	$k-4w+4+2kw^2-4kw+2k-7w^2+14w-7=0$	A1
	$(2k-7)x^2-2(2k-5)x+3(k-1)=0$	A1 (8) **[12]**

Type MS · The discriminant: equal, real or no real roots

Answer 3 of 12 · Question 7 part a · 4 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

	Notes
(a)	
M1	For use of $b^2 - 4ac$ (Ignore inequality for this mark)
M1	For a 3 TQ ≤ 0 oe
M1	For solving their 3TQ (Ignore inequality for this mark) May be implied by $-\frac{1}{2}$ and 4 seen as critical values
A1	For $-\frac{1}{2} \leq k \leq 4$ Allow $<$ instead of $\leq$ Accept $-\frac{1}{2} < k < 4$
(b)	
B1	For $\alpha + \beta = \frac{4}{k}$ and $\alpha\beta = \frac{2k-7}{k}$
M1	For $\frac{\alpha+1}{\alpha} + \frac{\beta+1}{\beta} = \frac{2\alpha\beta+\alpha+\beta}{\alpha\beta}$
M1	For substitution into $\frac{2\alpha\beta+\alpha+\beta}{\alpha\beta}$ with some attempt to simplify
A1	For $\frac{2(2k-5)}{2k-7}$
M1	For $\frac{\alpha+1}{\alpha} \times \frac{\beta+1}{\beta} = \frac{\alpha\beta+\alpha+\beta+1}{\alpha\beta}$
M1	For substitution into $\frac{\alpha\beta+\alpha+\beta+1}{\alpha\beta}$ with some attempt to simplify
A1	For $\frac{3(k-1)}{2k-7}$
A1	For $(2k-7)x^2 - 2(2k-5)x + 3(k-1) = 0$
	Alternative
B1	For $w = \frac{x+1}{x}$
M1	For rearranging to make x the subject
M1	For substitution of $x = \frac{1}{w-1}$ into the quadratic
A1	For $\frac{k}{(w-1)^2} - \frac{4}{w-1} + 2k - 7 = 0$
M1	For multiplying by $(w-1)^2$
M1	For expanding brackets
A1	For $k - 4w + 4 + 2kw^2 - 4kw + 2k - 7w^2 + 14w - 7 = 0$
A1	For $(2k-7)x^2 - 2(2k-5)x + 3(k-1) = 0$

Type MS · The discriminant: equal, real or no real roots

Answer 3 of 12 · Question 7 part a · 4 marks

2020 NOV · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · The discriminant: equal, real or no real roots

Answer 4 of 12 · Question 1 part whole · 6 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part whole in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0,\quad q\neq 0$ leading to $x=....$

Method marks for differentiation and integration:

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Use of a formula:

Generally, the method mark is gained by **either**

Type MS · The discriminant: equal, real or no real roots

Answer 4 of 12 · Question 1 part whole · 6 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part whole in this mark scheme extract

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

or, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working:

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."

Exact answers:

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark

Type MS · The discriminant: equal, real or no real roots

Answer 4 of 12 · Question 1 part whole · 6 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part whole in this mark scheme extract

Paper 1		
Question number	Scheme	Marks
1	$b^2 - 4ac \geq 0$	
	$(k+5)^2 - 4k(3k+6) \geq 0$	M1
	$k^2 + 10k + 25 - 12k^2 - 24k \geq 0$	M1
	$11k^2 + 14k - 25 \leq 0$	A1
	$(11k + 25)(k - 1) \leq 0$	M1
	[Critical values are $-\frac{25}{11}$ and 1]	
	$-\frac{25}{11} \leq k \leq 1$ oe	M1A1
		Total 6 marks

Type MS · The discriminant: equal, real or no real roots

Answer 5 of 12 · Question 1 part whole · 4 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part whole in this mark scheme extract

1	$[\pm(2k+6)]^2-4\times k\times 16$ $4k^2-40k+36=0$ $\rightarrow\{4\}(k-1)(k-9)\{=0\}\rightarrow k=...$ $k=1,9$	M1 dM1 M1 A1 [4]
		Total 4 marks

Mark	Notes
M1	**Allow working on expression, equation or any inequality for all M marks** Applies $b^2-4ac=[\pm(2k+6)]^2-4\times k\times 16$ and might be seen embedded in an attempt at the quadratic formula.
dM1	Forms a 3TQ in k by attempting to expand and collect the like terms.
M1	Attempts to solve their 3TQ in k to find at least one real value of k by any correct method (See General Principles). Condone labelling their $\boldsymbol{k}$ as $\boldsymbol{x}$ for this method mark. If there is no method shown, (use of a calculator) then the 3TQ must be correct and both k values must be fully correct ($k = 1$, $k = 9$) for evidence for this mark.
A1	For $k = 1$, 9 selected as their final answers. (Extra inequality answers for k must be rejected)

Question	Scheme	Marks
1 **ALT I** **(Via completing Square)**	$k(x-\frac{k+3}{k})^2-\frac{(k+3)^2}{k}+16$ $16-\frac{(k+3)^2}{k}=0\rightarrow k^2-10k+9=0$ $(k-1)(k-9)\{=0\}\rightarrow k=...$ $k=1,9$	M1 dM1 M1 A1 [4]
		Total 4 marks

Mark	Notes
M1	Attempts to complete the square for $kx^2-(2k+6)x+16$ (See General Principles) Similarly, divides both sides of the given quadratic equation by k first, then attempts to complete the square: $(x-\frac{k+3}{k})^2-\frac{(k+3)^2}{k^2}+\frac{16}{k}$
dM1	Sets the y coordinate of their turning point to be zero (= 0 can be implied) attempts to expand and collect like terms.
M1	Solves their 3TQ or polynomial to find at least one real value of k by any correct method (See General Principles) If there is no method shown, (use of a calculator) then their 3TQ must be correct and both k values must be fully correct ($k = 1$, $k = 9$)for evidence for this mark.
A1	For $k = 1$, 9 only

Type MS · The discriminant: equal, real or no real roots

Answer 5 of 12 · Question 1 part whole · 4 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks
1 ALT II (Via sum and product of roots)	$\alpha+\beta=\pm\frac{2k+6}{k},\alpha\times\beta=\frac{16}{k}\rightarrow\left\{\alpha+\alpha=\pm\frac{2k+6}{k},\alpha\times\alpha=\frac{16}{k}\right\}$ $(\frac{k+3}{k})^2=\frac{16}{k}\rightarrow k^2-10k+9=0$ $(k-1)(k-9)\{=0\}\rightarrow k=...$ $k=1\ ,\ 9$	M1 dM1 M1 A1 [4]
		Total 4 marks

Mark	Notes
M1	For: $\alpha+\beta=\pm\frac{2k+6}{k},\alpha\times\beta=\frac{16}{k}\rightarrow\left\{\alpha+\alpha=\pm\frac{2k+6}{k},\alpha\times\alpha=\frac{16}{k}\right\}$
dM1	Uses the fact that the quadratic equation has equal roots $\alpha=\beta$, sets up a correct equation in k, expands, collects like terms to form a 3TQ (or a Cubic equation) e.g. $\left(\frac{2k+6}{2k}\right)^2=\frac{16}{k}\rightarrow 4k^3-40k^2+36k=0$ or $4k^2-40k+36=0$
M1	Solves their 3TQ or Cubic equation to find at least one real value of k by any correct method (See General Principles) If there is no method shown, (use of a calculator) then their 3TQ or Cubic must be correct and both k values must be fully correct ($k=1, k=9$)for evidence for this mark.
A1	For $k=1, 9$ only

ALT III: (**Via differentiation**)

M1: Differentiates $kx^2-(2k+6)x+16$ with respect to x to achieve a linear expression

dM1: Sets their derivative to 0, solves for x and substitutes x back to $kx^2-(2k+6)x+16=0$ to form a 3TQ.

M1A1 same as ALT II

e.g. $kx^2-(2k+6)x+16\rightarrow 2kx-2k-6=0\rightarrow x=1+\frac{3}{k}$

$\rightarrow k\left(1+\frac{3}{k}\right)^2-(2k+6)(1+\frac{3}{k})+16=0$

$\rightarrow k=...$

Send to review if not sure

Type MS · The discriminant: equal, real or no real roots

Answer 5 of 12 · Question 1 part whole · 4 marks

2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · The discriminant: equal, real or no real roots

Answer 6 of 12 · Question 2 part whole · 5 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part whole in this mark scheme extract

number		
2	$(p-1)^2 - 4\times 2\times -2p > 0 \Rightarrow p^2 + 14p + 1 > 0$	M1A1
	critical values	
	$p = \dfrac{-14 \pm \sqrt{14^2 - 4\times 1\times 1}}{2} = -7 \pm 4\sqrt{3}$	M1
	$p < -7 - 4\sqrt{3}$ OR $p > -7 + 4\sqrt{3}$	M1A1 [5]
		Total 5 marks

Type MS · The discriminant: equal, real or no real roots

Answer 6 of 12 · Question 2 part whole · 5 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · The discriminant: equal, real or no real roots

Answer 7 of 12 · Question 1 part whole · 4 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks
1	$b^2-4ac=(k+8)^2-4\times2\times k=\left[k^2+8k+64\right]$	M1
	$k^2+8k+64=(k+4)^2-16+64=(k+4)^2+48$	dM1A1
	Conclusion: Irrespective of the value of k, $(k+4)^2\geqslant 0$ and so $b^2-4ac>0$ therefore the equation will always have distinct (real) roots.	A1 [4]
		Total 4 marks

Mark	Notes
M1	Applies the discriminant on the given QE Ignore $=0,\quad \geqslant 0,\quad >0$ This must be correct for this mark $\left[(k+8)^2-4\times2\times k\right]$. Ignore subsequent simplification errors.
dM1	Completes the square **ONLY** on **their** expression for the discriminant provided it is a 3TQ The expression must be of the form $(k+4)^2\pm X$ where $X\neq0, 64$ **NB:** This mark is dependent on the first M mark. Use of calculus to find $k=-4$ is M0
A1	For the correct expression.
A1	For a suitable conclusion. Only accept for this final mark a conclusion as follows [or an equivalent]. $(k+4)^2\geqslant 0$ and so $(k+4)^2+48>0$ hence it will always have **(real), distinct** roots. Note: $(k+4)^2+48\geqslant 0$ is A0

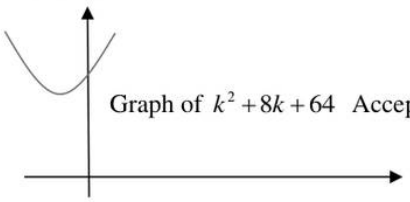
Type MS · The discriminant: equal, real or no real roots

Answer 7 of 12 · Question 1 part whole · 4 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 3 · find part whole in this mark scheme extract

Alternative methods for last 3 marks.	
ALT 1 finds discriminant of $k^2+8k+64$	
M1	Finds discriminant of **their** $k^2+8k+64 \Rightarrow D=64-4\times1\times64=-192$
A1	Obtains the correct expression in terms of k with the correct value of D
A1	Conclusion: $D<0$ so $k^2+8k+64$ does not have real roots. k^2 is positive so we know it has a minimum so $k^2+8k+64>0$ This means that $k^2+8k+64>0$ and therefore the discriminant will be > 0 meaning that the equation has distinct real roots for all values of k.

ALT 2 – Uses calculus	
M1	$y=k^2+8k+64 \Rightarrow \frac{dy}{dk}=2k+8=0$ at max/min
A1	So coordinates of max/min are $(-4,\ 48)$ and finds second derivative $\frac{d^2y}{dk^2}=2$
A1	Conclusion: $\frac{d^2y}{dk^2}=2>0$ hence it is a minimum. $(-4,\ 48)$ is the minimum point and so $k^2+8k+64>0$ This means that $k^2+8k+64>0$ and therefore the discriminant will be > 0 meaning that the equation has distinct real roots for all values of k.

ALT 3 – Uses a graph	
M1	Draws a sketch.  Graph of $k^2+8k+64$ Accept anywhere in positive y
A1	The graph is placed correctly with the min point in the 4th quadrant.
A1	Conclusion: the graph is always positive with its min point as shown in the diagram. This means that $k^2+8k+64>0$ and therefore the discriminant will be > 0 meaning that the equation has distinct real roots for all values of k.

Type MS · The discriminant: equal, real or no real roots

Answer 7 of 12 · Question 1 part whole · 4 marks

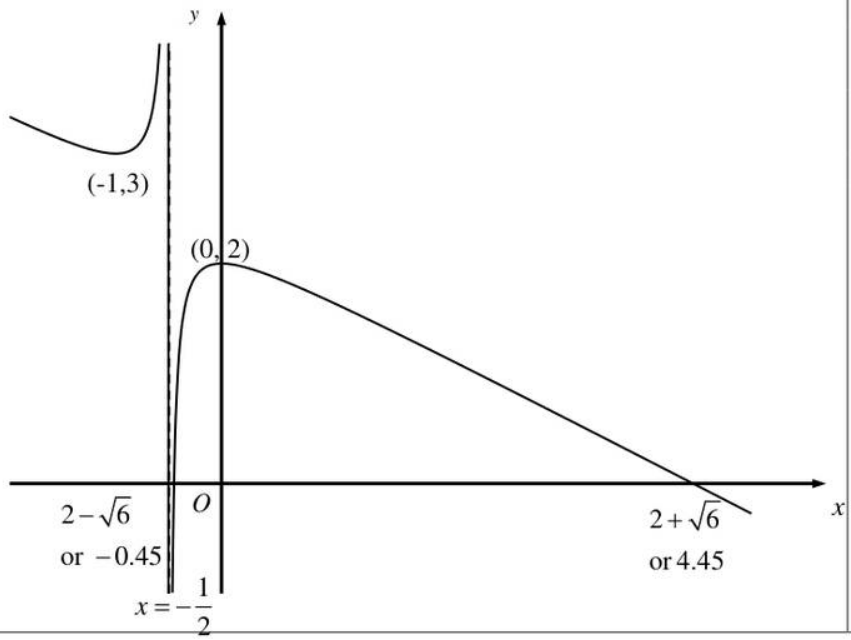
2023 JUN · Paper 02 (regional R) · extract 3 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · The discriminant: equal, real or no real roots

Answer 8 of 12 · Question 9 part b · 4 marks

2020 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

9(a)	$y=\dfrac{2+4x-x^2}{2x+1} \Rightarrow x^2-4x-2+2yx+y\ (=0)$ $x^2+(2y-4)x+(y-2)=0$	M1A1A1 (3)
(b)	$(2y-4)^2 \geq 4(y-2)$ $4y^2-16y+16=4y-8 \Rightarrow 4y^2-20y+24=0$ $y \leq 2$ or $y \geq 3$ *	M1 M1A1 A1cso (4)
(c)	$y=\dfrac{2+4x-x^2}{2x+1}$ $\dfrac{dy}{dx}=\dfrac{(4-2x)(2x+1)-2(2+4x-x^2)}{(2x+1)^2}$ **See notes for product rule method** $\dfrac{dy}{dx}=0 \Rightarrow (4-2x)(2x+1)-2(2+4x-x^2)=0$ $2x(x+1)=0 \Rightarrow x=0,\ -1$ stationary points are $(0,2)\ (-1,3)$	M1A1A1 M1 A1 A1 (6)
ALT	$x^2+(2y-4)x+(y-2)=0 \Rightarrow 2x+(2y-4)+2\dfrac{dy}{dx}x+\dfrac{dy}{dx}=0$ $\dfrac{dy}{dx}=0 \Rightarrow x+y=2$ (using (b)) stationary points are $(0,2)\ (-1,3)$	M1A1A1 M1 A1A1
(d)		B1 curve (i) M1A1 (M1 finding coords, A1 correct (oe or min 2dp) and on diagram (ii) B1 (iii) B1ft (5) [18]

Type MS · The discriminant: equal, real or no real roots

Answer 8 of 12 · Question 9 part b · 4 marks

2020 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

(a)M1	Re-write the equation of C without fractions and rearrange to the required form.
A1	Correct value for a and either b or c. These values need not be stated explicitly.
A1	Fully correct equation. Values of a, b and c need not be stated explicitly. Condone missing brackets round "$y-2$". Award this mark when the equation is reached – isw any listing of values with incorrect signs.
(b)	
M1	Use "$b^2 \geqslant 4ac$" for their equation
M1	Re-arrange their inequality to a 3TQ in y. Allow an equation here.
A1	Correct 3TQ, as shown or any equivalent
A1cso	Deduce the CVs (need not be shown explicitly) and state the (given) inequalities. There must be no errors in the working but the equation, if correct, can be solved easily so no working need be shown. Condone use of "and" instead of "or".
(c)	
M1	Differentiate the equation of C using the quotient rule. The denominator must be correct and the numerator must consist of the difference of 2 terms of the type shown. The product rule can be used.
A1	Either numerator term correct
A1	Fully correct numerator
ALT:	**Product Rule** $y=\left(2+4x-x^2\right)\left(2x+1\right)^{-1}$
M1A1 **A1**	$\frac{\mathrm{d}y}{\mathrm{d}x}=\left(4-2x\right)\left(2x+1\right)^{-1}-2\left(2+4x-x^2\right)\left(2x+1\right)^{-2}$ M1: rewrite without denominator and attempt product rule. Difference of 2 terms of the form shown needed (Difference because of the negative power) A1 Either term correct A1 Second term correct
M1	Equate the numerator of their derivative to 0 and solve to x = ...(any valid method of solving a quadratic with 2 or 3 terms) If product rule used must multiply through by $\left(2x+1\right)^2$
A1	Both x values correct
A1	Both stationary points correct
ALT	
M1	Use implicit differentiation on the re-arranged equation
A1	Correct derivative of $\left(2y-4\right)x$ (inc use of product rule)
A1	Fully correct derivative
M1	Set $\frac{\mathrm{d}y}{\mathrm{d}x}=0$ and use the result from (b) to obtain solutions
A1	One correct stationary point
A1	Both correct stationary points
(d)	
B1	Shape: Two parts, one above $y = 3$ and the other below $y = 2$
(i)M1	Attempt to find the x coordinates of the crossing points
A1	Correct coordinates shown on their sketch, 2 crossing points only. $y = 0$ need not be seen. There must be a curve through these points.
(ii)B1	The asymptote must be drawn and labelled (by its equation or by showing the x coordinate of the point where it crosses the x-axis). There must be at least one part of the curve which is asymptotic to the line $x=-\frac{1}{2}$. No part of the curve should touch/cross the asymptote or curve dramatically away from the line.
(iii)B1ft	Label the stationary points with their coords. Follow through provided the result is sensible.

Type MS · The discriminant: equal, real or no real roots

Answer 8 of 12 · Question 9 part b · 4 marks

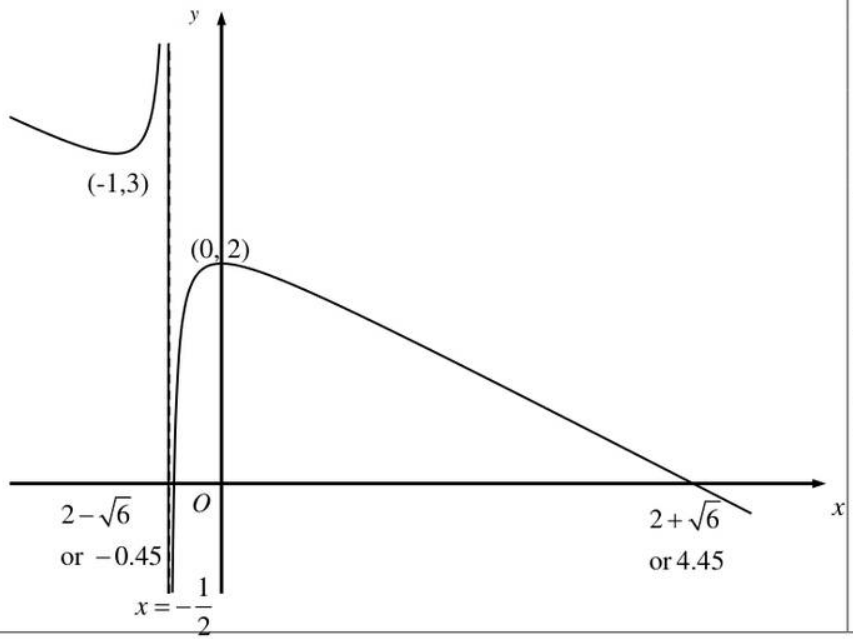
2020 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question Number	Scheme	Marks

Type MS · The discriminant: equal, real or no real roots

Answer 9 of 12 · Question 9 part a · 3 marks

2020 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

9(a)	$y=\dfrac{2+4x-x^2}{2x+1} \Rightarrow x^2-4x-2+2yx+y\ (=0)$ $x^2+(2y-4)x+(y-2)=0$	M1A1A1 (3)
(b)	$(2y-4)^2 \geq 4(y-2)$ $4y^2-16y+16=4y-8 \Rightarrow 4y^2-20y+24=0$ $y \leq 2$ or $y \geq 3$ *	M1 M1A1 A1cso (4)
(c)	$y=\dfrac{2+4x-x^2}{2x+1}$ $\dfrac{dy}{dx}=\dfrac{(4-2x)(2x+1)-2(2+4x-x^2)}{(2x+1)^2}$ **See notes for product rule method** $\dfrac{dy}{dx}=0 \Rightarrow (4-2x)(2x+1)-2(2+4x-x^2)=0$ $2x(x+1)=0 \Rightarrow x=0,\ -1$ stationary points are $(0,2)\ (-1,3)$	M1A1A1 M1 A1 A1 (6)
ALT	$x^2+(2y-4)x+(y-2)=0 \Rightarrow 2x+(2y-4)+2\dfrac{dy}{dx}x+\dfrac{dy}{dx}=0$ $\dfrac{dy}{dx}=0 \Rightarrow x+y=2$ (using (b)) stationary points are $(0,2)\ (-1,3)$	M1A1A1 M1 A1A1
(d)		B1 curve (i) M1A1 (M1 finding coords, A1 correct (oe or min 2dp) and on diagram (ii) B1 (iii) B1ft (5) [18]

Type MS · The discriminant: equal, real or no real roots

Answer 9 of 12 · Question 9 part a · 3 marks

2020 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

(a)M1	Re-write the equation of C without fractions and rearrange to the required form.
A1	Correct value for a and either b or c. These values need not be stated explicitly.
A1	Fully correct equation. Values of a, b and c need not be stated explicitly. Condone missing brackets round "$y-2$". Award this mark when the equation is reached – isw any listing of values with incorrect signs.
(b)	
M1	Use "$b^2 \geqslant 4ac$" for their equation
M1	Re-arrange their inequality to a 3TQ in y. Allow an equation here.
A1	Correct 3TQ, as shown or any equivalent
A1cso	Deduce the CVs (need not be shown explicitly) and state the (given) inequalities. There must be no errors in the working but the equation, if correct, can be solved easily so no working need be shown. Condone use of "and" instead of "or".
(c)	
M1	Differentiate the equation of C using the quotient rule. The denominator must be correct and the numerator must consist of the difference of 2 terms of the type shown. The product rule can be used.
A1	Either numerator term correct
A1	Fully correct numerator
ALT:	**Product Rule** $y=\left(2+4x-x^2\right)\left(2x+1\right)^{-1}$
M1A1 **A1**	$\frac{\mathrm{d}y}{\mathrm{d}x}=\left(4-2x\right)\left(2x+1\right)^{-1}-2\left(2+4x-x^2\right)\left(2x+1\right)^{-2}$ M1: rewrite without denominator and attempt product rule. Difference of 2 terms of the form shown needed (Difference because of the negative power) A1 Either term correct A1 Second term correct
M1	Equate the numerator of their derivative to 0 and solve to x = ...(any valid method of solving a quadratic with 2 or 3 terms) If product rule used must multiply through by $\left(2x+1\right)^2$
A1	Both x values correct
A1	Both stationary points correct
ALT	
M1	Use implicit differentiation on the re-arranged equation
A1	Correct derivative of $\left(2y-4\right)x$ (inc use of product rule)
A1	Fully correct derivative
M1	Set $\frac{\mathrm{d}y}{\mathrm{d}x}=0$ and use the result from (b) to obtain solutions
A1	One correct stationary point
A1	Both correct stationary points
(d)	
B1	Shape: Two parts, one above $y = 3$ and the other below $y = 2$
(i)M1	Attempt to find the x coordinates of the crossing points
A1	Correct coordinates shown on their sketch, 2 crossing points only. $y = 0$ need not be seen. There must be a curve through these points.
(ii)B1	The asymptote must be drawn and labelled (by its equation or by showing the x coordinate of the point where it crosses the x-axis). There must be at least one part of the curve which is asymptotic to the line $x=-\frac{1}{2}$. No part of the curve should touch/cross the asymptote or curve dramatically away from the line.
(iii)B1ft	Label the stationary points with their coords. Follow through provided the result is sensible.

Type MS · The discriminant: equal, real or no real roots

Answer 9 of 12 · Question 9 part a · 3 marks

2020 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question Number	Scheme	Marks

Type MS · The discriminant: equal, real or no real roots

Answer 10 of 12 · Question 5 part a · 6 marks

2023 JAN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

5 a	$\left(y=1+\frac{k}{2}x\Rightarrow\right)$ eg $kx^2-x\left(1+\frac{k}{2}x\right)+(k+1)x-1=0$ or eg $\frac{kx^2+(k+1)x-1}{x}=\frac{k}{2}x+1$	M1
	$kx^2-x-\frac{k}{2}x^2+kx+x-1=0$ or $2kx^2+2(k+1)x-2=kx^2+2x$ oe	M1
	And an attempt to simplify must be made.	
	$kx^2+2kx-2=0$ oe	A1
	$\left(b^2-4ac=0\Rightarrow\right)("2k")^2-4("k")("-2")=0$ or $(x+1)^2=1+\frac{2}{k}$	M1
	$4k(k+2)=0$ or $k+2=0$ or $1+\frac{2}{k}=0$	dM1
	$k=-2$	A1 (6)
ALT **Final 3 marks**	Realises if the line intersects the curve only once, it is a tangent at point A – final 3 marks	
	$y=\frac{kx^2+(k+1)x-1}{x}=\left(kx+k+1-\frac{1}{x}\right)$ and $\left(\frac{dy}{dx}=\right)\frac{k}{2}$	M1
	$\left(\frac{dy}{dx}=\right)k(+0)+\frac{1}{x^2}=\frac{k}{2}\Rightarrow x^2=-\frac{2}{k}$	
	$k\left(-\frac{2}{k}\right)+2k\sqrt{-\frac{2}{k}}-2=0\Rightarrow 2k\sqrt{-\frac{2}{k}}=4\Rightarrow 4k^2\left(-\frac{2}{k}\right)=16\Rightarrow -8k=16$	dM1
	$k=-2$	A1
b	When $k=-2$ $-2x^2-4x-2=0$ or $(x+1)^2=1+\frac{2}{-2}$	
	$(x+1)^2=0$	M1
	$x=-1\Rightarrow y=2$ So $(-1,2)$	A1 (2)
		Total 8 marks

Type MS · The discriminant: equal, real or no real roots

Answer 10 of 12 · Question 5 part a · 6 marks

2023 JAN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For correctly substituting $y=1+\frac{k}{2}x$ into $kx^2-xy+(k+1)x=1$
	M1	For a complete expansion of $x\left(1+\frac{k}{2}x\right)$ and $(k+1)x$. Allow one error in the expansion. For multiplying the equation throughout by $2x$ or $-2x$, x or $-x$ to eliminate any algebraic denominators, allow one processing error. There must be an attempt to simplify follow either of these steps. If any other manipulation is seen, full marks can be awarded if the steps lead to a correct equation or a three term quadratic with one processing error made. If there is relevant work, not leading to either of these, please send to review.
	A1	For a correct three term quadratic. oe
	M1	For **clear and correct** use of $b^2-4ac=0$ or a **full and correct** completing the square process, using their expressions for a, b and c. $a,b,c\neq 0$
	dM1	For factorising or any attempt to solve their quadratic in k, or setting their $1+\frac{2}{k}=0$ See general guidance for acceptable attempt to solve a quadratic. For the case of a 2TQ with terms in k and k^2 (which a fully correct solution delivers), we need to see a correct factorisation in the case of this mark or, as the candidate is told k is no zero, they may also correctly divide throughout by k. Dependent on the previous method mark. Allow this mark to be implied if k is correct. Otherwise, the method must be shown to gain this mark.
	A1	$k=-2$, must dismiss $k=0$, if found.
ALT	**States and use the gradient of C is $\frac{k}{2}$.**	
	M1	As main scheme.
	M1	As main scheme.
	A1	As main scheme.
	M1	Differentiates C and puts the gradient = $\frac{k}{2}$ to get an equation of the form $mk+\frac{n}{x^2}=\frac{k}{2}$ $m,n\neq 0$ (doesn't need to be simplified) but must lead to an expression of the form $x^2=-\frac{l}{k}$ $l\neq 0$
	dM1	Substitutes their expression for x^2 into their quadratic equation and rearranges to give an expression of the form $-pk=q$ or $pk=-q$ $p>0,k>0$
	A1	$k=-2$ (must be the only solution given).
(b)	M1	For correct substitution of their value of k into their 3 term quadratic from part a and a minimally acceptable attempt to solve for x. See general guidance.
	A1	For $(-1,2)$ or $x=-1$ and $y=2$

Type MS · The discriminant: equal, real or no real roots

Answer 10 of 12 · Question 5 part a · 6 marks

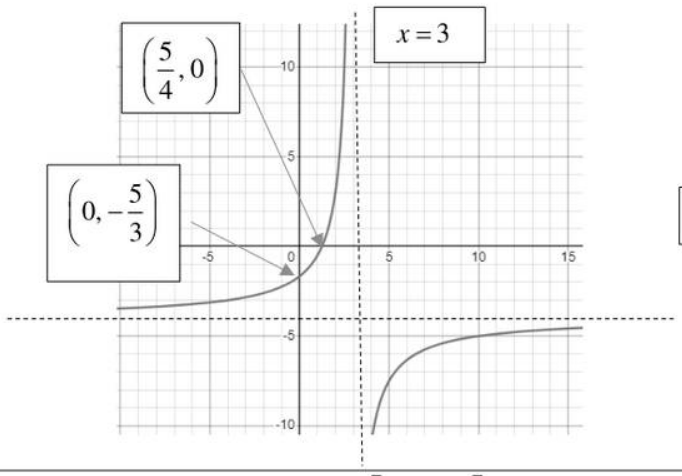
2023 JAN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks
	3	B1 B1

Type MS · The discriminant: equal, real or no real roots

Answer 11 of 12 · Question 10 part c · 9 marks

2024 JUN · Paper 01 · extract 1 of 5 · find part c in this mark scheme extract

10(a)	$\left(ax-5=0,\ x=\frac{5}{4}\Rightarrow\right)a=4$ $\left(\text{Line parallel to the } y\text{-axis is } x=3\Rightarrow\right)b=3$	B1 B1 {2}
(b)		B1 B1ftB1 B1ftB1ft [5]
(c)	$\left(\frac{dy}{dx}=\right)\frac{(3-x)(4)-(4x-5)(-1)}{(3-x)^2}=\left[\frac{7}{(3-x)^2}\right]$ (Gradient of l =) $\frac{7}{4}$ $\frac{7}{4}=\frac{7}{(3-x)^2}\Rightarrow(3-x)^2=4\Rightarrow 3-x=\pm 2\Rightarrow x=1,5$ When $x=1$, $y=-\frac{1}{2}$ and when $x=5$, $y=-\frac{15}{2}$ Equation of line when $x=1$: $y--\frac{1}{2}=\frac{7}{4}(x-1)\Rightarrow 4y-7x=-9$ Equation of line when $x=5$: $y--\frac{15}{2}=\frac{7}{4}(x-5)\Rightarrow 4y-7x=-65$ $\Rightarrow -65<k<-9$	M1 B1 M1A1 M1 A1 (B1B1 on ePen) ddM1 ddM1 A1 [9]

Type MS · The discriminant: equal, real or no real roots

Answer 11 of 12 · Question 10 part c · 9 marks

2024 JUN · Paper 01 · extract 2 of 5 · find part c in this mark scheme extract

ALT	$(y=)\dfrac{7x+k}{4}=\dfrac{"4"x-5}{"3"-x}$ oe $4\left(\dfrac{"4"x-5}{"3"-x}\right)-7x=k$ oe	M1 A1(B1 on ePen)
	$16x-20=(7x+k)("3"-x)$ oe	
	$16x-20=21x-7x^2+3k-kx \Rightarrow 7x^2+kx-5x-20-3k(=0)$ oe	M1 A1
	$(\Rightarrow 7x^2+(k-5)x-20-3k(=0))$ oe	
	$(k-5)^2-4(7)(-20-3k)(<0)$	M1A1 (B1 B1 on ePen)
	$k^2+74k+585(<0)$	
	$(k=)\dfrac{-74\pm\sqrt{(74)^2-4\times(1)\times585}}{2}$ or $(k-9)(k-65)(=0)$	ddM1 ddM1
	$-65<k<-9$	A1 [9]

Type MS · The discriminant: equal, real or no real roots

Answer 11 of 12 · Question 10 part c · 9 marks

2024 JUN · Paper 01 · extract 3 of 5 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the value of $a = 4$ or $b = 3$
	B1	For the value of $a = 4$ and $b = 3$
Note: These are independent of method marks, not given answers. Unless the answers come from **very obviously** incorrect working, these marks should be awarded for sight of the correct values.		
(b)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn and **must not** **obviously** 'bend back' on themselves. Mark intention.
	For the following marks, where candidates have used incorrect values of a and b, examiners will find using desmos.com or similar packages useful. Type into desmos $y = (ax - b)/(b - x)$ and add 'sliders' for a and b	
	B1ft	For the horizontal asymptote of $y = -4$ drawn in the correct place and labelled with its equation or where the line passes through the axis. There **must** be at least one branch of a reciprocal curve present which **must** not cross or **obviously** bend back from the asymptote(s) ignore any other curves present. Allow follow through **only** of $y = -$ (their a) and only if other conditions met.
	B1	For the vertical asymptote $x = 3$ drawn in the correct place. There **must** be at least one branch of a reciprocal curve present which **must** not cross or **obviously** bend back from the asymptote(s), ignore any other curves present.
	B1ft	A single curve (or even line) passing through $\left(0, -\frac{5}{3}\right)$ marked clearly on the graph as a coordinate or as a crossing point on the axis. Ignore any other branches present. Allow follow through only of $\left(0, -\frac{5}{\text{their } b}\right)$
	B1ft	A single curve (or even line) passing through both $\left(\frac{5}{4}, 0\right)$ and $\left(0, -\frac{5}{3}\right)$ both marked clearly on the graph as a coordinate or as crossing points on the axes. Ignore any other branches present. Allow follow through only of $\left(0, -\frac{5}{\text{their } b}\right)$

Type MS · The discriminant: equal, real or no real roots

Answer 11 of 12 · Question 10 part c · 9 marks

2024 JUN · Paper 01 · extract 4 of 5 · find part c in this mark scheme extract

(c)	M1	For applying the quotient (or product rule). • With two terms in the numerator subtracted either way around. • The differentiation on both terms must be correct. • The denominator must be squared. $\frac{(3-x)(4)-(4x-5)(-1)}{(3-x)^2}$ or $\frac{(4x-5)(-1)-(3-x)(4)}{(3-x)^2}$ (if using quotient rule) Or $4(3-x)^{-1} \pm (4x-5)(3-x)^{-2}$ (if using product rule)
	B1	For (Gradient of l =) $\frac{7}{4}$
	M1	For correctly setting their gradient of l = to their derivative and an attempt to solve to find 2 values of x. Allow one error in rearrangement. Their derivative must be of the form $\frac{rx+s}{(3-x)^2}$ s **or** r could be 0, but not both
	A1	For $x = 1$ **and** 5
	M1 (B1 on ePen)	For substitution of either of their x values into the equation of their curve.
	A1 (B1 on ePen)	For $y = -\frac{1}{2}$ **and** $y = -\frac{15}{2}$
	ddM1	For correctly forming an equation of the line for either pair of their values and their gradient and rearranging (allow errors in rearrangement) to the form $4y - 7x = k$. Dependent on 1st 2 method marks.
	ddM1	For correctly forming an equation of the line for both pairs of their values and their gradient and rearranging (allow rearrangement errors) to the form $4y - 7x = k$ Dependent on 1st 2 method marks.
	A1	For the correct inequality as shown.
ALT	M1	For rearrangement of l to the form $y = \frac{7}{4}x + c \quad c \neq 0$ and placing equal to their equation of C or substitution of the equation for curve C in the equation for l. Allow $4\left(\frac{"4"x-5}{"3"-x}\right) = k \pm 7x$ Condone using an inequality sign, follow through their a and b
	A1 (B1 on ePen)	For a correct unsimplified equation. **DO NOT** condone using an inequality sign
	M1	For an attempt to rearrange their equation of the form $mx + c = \frac{4x-5}{3-x}$ $m, c \neq 0$, to allow them to reach a "3TQ", allow one error in rearrangement. Condone using an inequality sign By "3TQ", we means terms in x, x^2 and constant, all on 'one side' even if the equals sign is missing or an incorrect inequality sign is shown, terms simplified where possible.
	A1	For the correct "3TQ" shown (oe), the term in x does not have to be factorised and = 0 may be omitted. "3TQ" defined as above
	M1 (B1 on ePen)	For the correct formulation of the discriminant from their "3TQ". Inequality sign doesn't need to be shown or can be incorrect.
	A1 (B1 on ePen)	For the correct unsimplified discriminant. Inequality sign doesn't need to be shown or can be correct.
	ddM1	For a correct method to solve their quadratic equation in k, leading to a value of k. Dependent on first 2 method marks. Mark awarded for the correct **method.**
	ddM1	For a correct method to solve their quadratic equation in k, leading to two values of k. Dependent on 1st 2 method marks. Mark awarded for the correct **method.**
	A1	For the inequality shown.

Type MS · The discriminant: equal, real or no real roots

Answer 11 of 12 · Question 10 part c · 9 marks

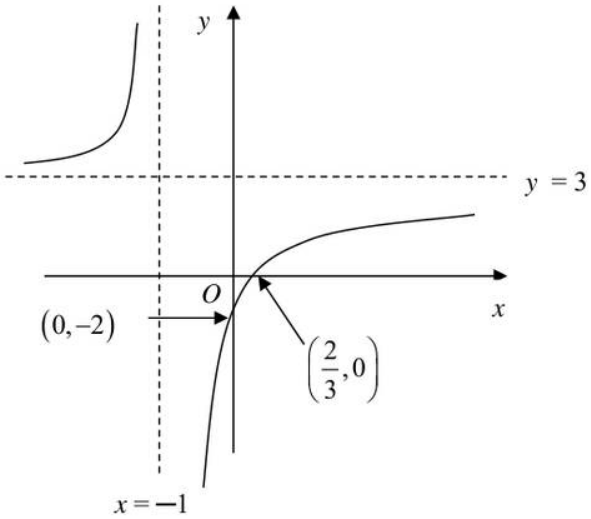
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Type MS · The discriminant: equal, real or no real roots

Answer 12 of 12 · Question 11 part d · 7 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 5 · find part d in this mark scheme extract

11 (a)	(i) $y=3$ (ii) $x=-1$	B1B1 [2]
(b) (i) (ii)	$0=\frac{3x-2}{x+1}\Rightarrow 3x=2\Rightarrow x=\frac{2}{3}$ $y=\frac{3\times 0-2}{0+1}=-2$	B1 B1 [2]
(c)		B1 (shape and position) B1 (asymptotes) B1 (intersections) [3]
(d)	$mx+4=\frac{3x-2}{x+1}\Rightarrow mx^2+x(m+1)+6=0$ $b^2-4ac<0\Rightarrow (m+1)^2-4\times m\times 6<0\Rightarrow m^2-22m+1<0$ $(m-11)^2-120=0\Rightarrow m=11\pm\sqrt{120}$ Defines region $11-2\sqrt{30}<m<11+2\sqrt{30}$	M1 M1 M1A1 M1 depM1A1 [7]
		Total 14 marks

Type MS · The discriminant: equal, real or no real roots

Answer 12 of 12 · Question 11 part d · 7 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 5 · find part d in this mark scheme extract

(a) (i)	
B1	$y = 3$
(a) (ii)	
B1	$x = -1$
(b) (i)	
B1	$x = \frac{2}{3}$
(b) (ii)	
B1	$y = -2$
(c)	
B1	Two curves in the correct quadrant (Condone poor shape as long as intention is clear)
B1	Two asymptotes drawn and correctly labelled (Allow –1 and 3 correctly labelled on axes to count as labelled). There must be at least one part of the curve drawn
B1	Intersections labelled
(d)	
M1	Equates C and l (Condone use of $\neq$ and $<$)
M1	Forms a 3TQ (Condone the use of $\neq$ and $<$) (allow 1 error in the expansion of correct brackets or 1 error in simplifying)
M1	Use of $b^2 - 4ac < 0$ to form a 3TQ
A1	Simplify to $m^2 - 22m + 1 < 0$
M1	Solving the 3TQ If the quadratic is incorrect then the method for solving must be shown
depM1	Simplifying to the form $a \pm b\sqrt{2}$ Dependant on previous M1
A1	$11 - 2\sqrt{30} < m < 11 + 2\sqrt{30}$ (Allow $a = 11, b = 30$)

Type MS · The discriminant: equal, real or no real roots

Answer 12 of 12 · Question 11 part d · 7 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 5 · find part d in this mark scheme extract

Type MS · The discriminant: equal, real or no real roots

Answer 12 of 12 · Question 11 part d · 7 marks

2020 JAN · Paper 02 (regional R) · extract 4 of 5 · find part d in this mark scheme extract

Type MS · The discriminant: equal, real or no real roots

Answer 12 of 12 · Question 11 part d · 7 marks

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